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TEMPO Data Products, Science, and Applications

Xiaohang Shi and Yulong Fan contributed equally to this work.

Key Points:

- A physics-guided feature selection ML framework was developed to detect clouds from hyperspectral environmental satellite observations
- The PGML framework captures the diurnal evolution of cloud probability and spatial patterns across different seasons and surface types
- Validation against reference cloud masks shows high accuracy for TEMPO (BOA = 84%) and strong generalization to GEMS (BOA = 80%)

Supporting Information:

Supporting Information may be found in the online version of this article.

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A Physics-Guided Feature Selection Machine Learning Approach for Cloud Probability Retrieval and Detection From TEMPO Imagery

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Abstract Accurate cloud detection is critical for reliable retrievals of aerosols and trace gases from remote-sensing measurements. TEMPO provides unprecedented hourly observations over North America, but radiometric calibration biases, thousands of spectral channels, and limited spectral coverage make cloud detection challenging. Here, we develop a physics-guided feature selection machine learning (PGML) framework that integrates radiative transfer simulations with a LightGBM architecture to detect clouds from TEMPO hyperspectral observations. Our approach starts with radiative transfer simulations to identify the spectral channels most sensitive to clouds among the 2,056 available. An explainable AI method is then applied to rank feature importance, further refining model inputs to 25 ultraviolet and visible channels plus geometric angles, improving interpretability and computational efficiency. Using mature GOES-R cloud masks as reference labels, the model is trained and optimized with a cross-entropy loss function to predict cloud probability. The approach effectively captures diurnal variations in both cloud fraction and spatial structures. The resulting cloud masks show high classification performance against the reference masks, with a balanced overall accuracy (BOA) of 84.43%, outperforming the Level 1B cloud mask (73.86%), which tends to underestimate cloud cover. Independent validation with AERONET observations further confirms this improvement (BOA: 78.54% vs. 54.88%). Moreover, the framework generalizes well to GEMS, achieving BOA scores of 80.38%, 71.98%, and 77.42% against Himawari cloud masks, AERONET, and CALIPSO observations, respectively. These results highlight the framework's strong adaptability and transferability, as well as its potential to enhance atmospheric composition monitoring with the next generation of hyperspectral environmental satellites.

Plain Language Summary Clouds strongly affect satellite observations and can limit the retrieval of aerosols and trace gases. TEMPO provides hourly observations over North America, but its complex hyperspectral measurements make cloud detection challenging. We develop a physics-informed machine learning approach to improve cloud detection from TEMPO observations. Physical simulations are first used to identify the spectral wavelengths most sensitive to cloud presence, and machine learning is then applied to refine the input features and estimate cloud probability using selected spectral channels and viewing geometry, with mature cloud masks used as reference labels. Evaluation against existing satellite products and independent ground-based observations shows that the new method detects clouds effectively and better captures the diurnal evolution of cloud probability, particularly for thin clouds. The approach also performs well when applied to the GEMS satellite over Asia, demonstrating strong transferability across sensors and regions. Overall, this work provides a practical and reliable method for improving cloud detection from hyperspectral satellites, supporting more accurate satellite-based monitoring of atmospheric composition.

1. Introduction

Spaceborne remote sensing instruments provide extensive observations for continuous monitoring of the Earth's atmosphere and surface. However, a large portion of these observations is affected by clouds, which cover

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roughly 66% of the Earth's surface, according to the International Satellite Cloud Climatology Project (ISCCP) (Zhang et al., 2004). Clouds modify the paths of sunlight between the Sun and the satellite through anisotropic scattering, shielding the atmosphere and surface below while enhancing radiative interactions above (Harshvardhan & Corsetti, 1989; Hartmann et al., 1986; Zhao et al., 2014). Consequently, accurate cloud detection is essential for reducing uncertainties in a wide range of satellite-based applications, including aerosol retrievals, surface property mapping, and trace-gas quantification (Chand et al., 2009; Letu et al., 2012; Segal-Rozenhaimer et al., 2020; Sun et al., 2016).

Traditional cloud-detection algorithms typically exploit the higher reflectance and lower brightness temperatures of clouds relative to underlying surfaces. Early multispectral approaches, such as APOLLO (Saunders & Kriebel, 1988), ISCCP (Rossow & Garder, 1993), CO₂ slicing (Wylie & Menzel, 1989), and CLAVR (Stowe et al., 1991), distinguish cloudy from clear pixels using spectral thresholds in the visible, near-infrared, and thermal infrared domains, although such spectral contrasts don't always exist, such as in polar regions (Li & Leighton, 1991; Sun et al., 2017). These concepts also underpin operational algorithms for sensors such as MODIS (Ackerman et al., 1998), Landsat (Zhu & Woodcock, 2012; Zhu et al., 2015), and Geostationary Operational Environmental Satellite R (GOES-R) (Heidinger, 2010; Schmit et al., 2017). Threshold-based methods perform well for instruments with spectral bands that exhibit strong cloud–surface contrasts and accurate radiometric calibration. However, hyperspectral ultraviolet–visible sensors without longwave infrared coverage, primarily designed for trace-gas retrievals, often experience substantial radiometric calibration uncertainties. These uncertainties, together with the large number of spectral channels, pose challenges for conventional threshold-based algorithms that rely on precise radiance measurements. Despite these potential biases, hyperspectral measurements preserve their spectral structure, which can be leveraged to develop dedicated cloud-detection algorithms capable of accurately identifying clouds under diverse observing conditions.

Hyperspectral sensors such as the Ozone Monitoring Instrument (OMI) (Levelt et al., 2018) and the Tropospheric Monitoring Instrument (TROPOMI) (Veefkind et al., 2012) have demonstrated substantial value in atmospheric composition research, contributing to advances in cloud and aerosol retrievals as well as trace-gas monitoring (Landgraf et al., 2016; Li et al., 2024; Rao et al., 2022; Shi et al., 2024; Wang et al., 2024). To reduce the uncertainties introduced by cloud contamination, physics-based algorithms are commonly employed to identify and remove cloud-affected pixels from these data sets (Acarreta et al., 2004; Vasilkov et al., 2008, 2018; Veefkind et al., 2016). For example, the OMCLDO2 OMI cloud product retrieves effective cloud fraction and effective cloud pressure by applying a DOAS (differential optical absorption spectroscopy) fit to the O₂–O₂ absorption feature near 477 nm (Vasilkov et al., 2018). Veefkind et al. (2016) developed a similar OMI cloud algorithm based on the same O₂–O₂ absorption band, but employing an alternative spectral fitting approach and incorporating surface-geometry-dependent Lambertian-equivalent reflectivity (GLER). In contrast, TROPOMI extends cloud characterization into the near-infrared oxygen A-band (758–770 nm), treating clouds as scattering layers or variable cloud albedo and relying on O₂–A band measurements to improve retrieval performance (Dobber et al., 2006; Loyola et al., 2018; Veefkind et al., 2012). Despite their widespread use, these operational algorithms rely heavily on radiance contrasts and typically apply fixed thresholds for cloud-mask generation, which may limit their adaptability under complex surface and atmospheric conditions (Diego et al., 2021; Loyola & Rupprecht, 1998; Loyola et al., 2018).

The Tropospheric Emissions: Monitoring of Pollution (TEMPO) mission marks a major step forward, providing the first geostationary hyperspectral measurements of air quality over North America (Dang et al., 2025; Wang et al., 2025; Zoogman et al., 2017). Operating at hourly temporal resolution with high spatial and spectral coverage, TEMPO offers unique opportunities to monitor diurnal variations in emissions and atmospheric chemistry. Its development follows earlier geostationary missions such as the Geostationary Environment Monitoring Spectrometer (GEMS), launched in 2020 to monitor atmospheric composition over Asia (Kim et al., 2021; Kim et al., 2020). Like OMI, both GEMS and TEMPO rely on O₂–O₂ features for cloud characterization (Bo-Ram et al., 2024). TEMPO's operational cloud algorithm follows the O₂–O₂ approach adapted from OMI (Vasilkov et al., 2018). The TEMPO Level-2 CLDO4 product provides effective cloud fraction and cloud optical centroid pressure optimized for UV trace-gas retrievals (Wang et al., 2025). In addition, a separate cloud mask is generated from Level-1B (L1B) radiances, using three visible spectral bands (red, green, and blue) for true-color imagery. Alternative cloud masks can also be derived by thresholding the TEMPO effective cloud fraction. Nonetheless, both approaches retain limitations, particularly for aerosol retrievals, which require stringent cloud screening to minimize residual contamination.

Recent advances in machine learning (ML) demonstrate strong potential for improving cloud detection across diverse satellite platforms (Pang et al., 2023, 2026; Shang et al., 2024; Wang et al., 2022; Wei et al., 2020; Yang et al., 2022). ML models can leverage large data sets, incorporate multidimensional spectral and geometric features, and are less sensitive to calibration biases than traditional physical algorithms. Motivated by these capabilities, we develop a unified Physics-Guided Feature Selection Machine Learning (PGML) framework tailored to hyperspectral geostationary observations. The approach integrates radiative transfer simulations for physically grounded feature screening with eXplainable AI (XAI) for data-driven feature refinement. The final model combines selected spectral channels and geometric parameters within a LightGBM architecture trained using GOES-R cloud masks as supervisory labels. To further assess generalizability, we extend the framework to GEMS and evaluate performance using independent ground-based and satellite observations. This study presents the development, evaluation, and cross-sensor validation of this PGML framework for cloud probability retrieval and cloud-mask generation. The results demonstrate substantial improvements and highlight the method's robustness across seasons, surface types, illumination conditions, and different hyperspectral missions.

2. Data Source

2.1. TEMPO Imagery

The Tropospheric Emissions: Monitoring of Pollution (TEMPO) is NASA's first Earth Venture Instrument dedicated to monitoring air quality over North America from geostationary orbit (Zoogman et al., 2017). Hosted aboard the Intelsat-40e satellite and launched on 7 April 2023, TEMPO provides hourly daytime observations of key pollutants—including ozone (O_3), nitrogen dioxide (NO_2), and formaldehyde (HCHO)—at high spatial and spectral resolution, offering unprecedented capability to monitor diurnal variability in atmospheric composition (Wang et al., 2025).

Operating at $91^\circ W$, TEMPO is equipped with two charge-coupled device (CCD) detectors covering the ultra-violet (UV, ~ 293 – 494 nm) and visible (VIS, ~ 538 – 741 nm) spectral regions, with a combined total of 2,056 spectral channels and a spectral resolution of approximately 0.6 nm. The nadir spatial resolution is approximately 2 km (north–south) $\times$ 4.75 km (east–west) at the center of the field of regard, and the hourly revisit cycle enables continuous monitoring of dynamic atmospheric processes. This study utilizes the TEMPO V3 Level-1B (L1B) Geolocated Earth Radiances (RAD) product, which provides geolocated daytime Earth-shine radiances. UV and VIS radiances are stored in the NetCDF groups “band_290_490_nm” and “band_540_740_nm.”

2.2. Training and Validation Data Sets

2.2.1. GOES-R Cloud Products

To construct training data sets for cloud detection, we employ cloud masks from the GOES-R series. The GOES-R Advanced Baseline Imager (ABI), specifically GOES-East positioned at $-75.2^\circ W$, provides full-disk imagery every 5 min across 16 spectral bands spanning the visible, near-infrared, and infrared wavelengths (Zheng et al., 2023). Its frequent sampling facilitates efficient spatial and temporal collocation with TEMPO observations (McCorkel et al., 2020). We use the GOES-R Cloud Mask (CMA) product at a spatial resolution of 2 km, developed by the Nowcasting and Very Short Range Forecasting Satellite Application Facility (NWCSAF) (García-Pereda et al., 2019). The CMA algorithm applies a suite of multispectral threshold and contextual consistency tests that consider reflectance, brightness temperature, solar-view geometry, and atmospheric conditions to classify pixels as cloud-free, cloudy, or indeterminate (Karlsson et al., 2020). Its robust performance across daytime, nighttime, twilight, and sunglint conditions makes it an ideal supervisory reference for training machine learning models.

2.2.2. AERONET Observations

Independent validation is performed using cloud information inferred from the Aerosol Robotic Network (AERONET), a global network of ground-based sun photometers operating under a standardized calibration and data-processing framework (Holben et al., 1998). AERONET provides aerosol optical depth (AOD) at three processing levels: Level 1.0 (unscreened), Level 1.5 (cloud-screened), and Level 2.0 (quality-assured) (Giles et al., 2019). Cloud presence is inferred from discrepancies between unscreened and screened AOD measurements. For validation, TEMPO pixels are collocated spatially with AERONET sites and temporally within a ± 5 -

min window of the satellite overpass time. Level 1.0 and Level 1.5 data are used to identify cloudy and clear conditions, establishing independent ground-based benchmarks for algorithm performance. AERONET Level 1.5 filtering may occasionally remove optically thick aerosol plumes (e.g., wildfire smoke) alongside true clouds (Giles et al., 2019). However, such removal occurs only during very intense smoke events and represents a small fraction of the filtered data; most removed samples are still true clouds. More importantly, for TEMPO observations, AERONET represents the only independent validation source available to us in this study. It is not used as the primary reference for model training or evaluation, but rather as an independent benchmark.

2.2.3. TEMPO Cloud Products

For comparison, we collect both the TEMPO V3 official Level-2 CLDO4 cloud fraction product and the cloud mask provided in the Level 1B (L1B) data. The CLDO4 product retrieves effective cloud fraction using the 466 nm wavelength, while cloud optical centroid pressure is derived from the 477 nm O_2-O_2 absorption feature, employing an optimized spectral fitting window from 439 to 488 nm (Vasilkov et al., 2018; Wang et al., 2025). This retrieval relies on the TEMPO trace-gas spectral fitting framework applied to top-of-atmosphere radiances in the 477-nm O_2-O_2 absorption band and incorporates a climatology of GLER to reduce uncertainties from surface reflectance assumptions. The algorithm assumes a fixed cloud albedo of $A = 0.8$ and is specifically designed to support TEMPO trace-gas retrievals, ensuring internal consistency between CLDO4 and the trace-gas products. CLDO4 represents a radiatively coupled effective cloud fraction (assuming $A = 0.8$), while our output is a geometric cloud probability. The comparison is now framed as reflecting two distinct physical states rather than a direct bias.

In addition to radiances, the TEMPO L1B product includes a “cloud_mask_group”, derived from radiances in the red, green, and blue channels, along with scaled true-color red, green, and blue values generated using an RGB-based cloud classifier scheme. This classifier uses three selected spectral bands that approximately correspond to the “red”, “green”, and “blue” channels, and is based on the premise that clouds generally appear white and bright in satellite observations, in contrast to most underlying surface features (Chong et al., 2026). The TEMPO L1B cloud mask is derived primarily using threshold tests on visible reflectance, which is conceptually more similar to traditional optical cloud fraction retrievals. This differs fundamentally from the O_4 (or O_2-O_2) and O_2A band cloud retrievals, which provide radiometric (effective) cloud fractions that measure the amount of radiance attributed to clouds.

3. Methodology

3.1. Cloud Probability Retrieval Framework

To enable accurate cloud detection from hyperspectral geostationary sensors, we propose a Physics-Guided Feature Selection Machine Learning (PGML) framework that integrates radiative transfer simulations with a gradient boosting LightGBM model. The framework consists of four main steps: feature selection, model construction, cloud detection, and comparison and validation (Figure 1). Radiative transfer simulations provide physical insights into the spectral contrast between cloudy and clear conditions, while feature importance analysis ensures that the selected wavelengths are empirically effective for classification. Using the selected TEMPO spectral bands and geometric parameters, the LightGBM model is trained with GOES-R reference cloud masks as supervisory labels. The model outputs cloud probability, interpreted as cloud probability, and the final cloud mask is generated by applying an appropriate threshold.

3.1.1. Feature Selection

TEMPO contains 2,056 spectral channels, making the efficient selection of physically meaningful input features essential for machine learning applications. To filter suitable input bands for the machine learning model of a hyperspectral sensor, the radiative transfer model SCIATRAN (Mei et al., 2023; Rozanov et al., 2014) was used to simulate radiances under representative clear-sky and cloudy conditions, explicitly accounting for radiative contributions from clouds and the underlying surface. The simulations were performed using the default SCIATRAN configuration, employing plane-parallel scattering with atmospheric refraction enabled (reference wavelength: 550 nm) and a maximum refraction height of 35 km, under a representative daytime geometry with a solar zenith angle of 0° . No atmospheric trace gases were included, as the simulations were solely intended to assess relative radiance contrasts between cloudy and clear-sky conditions for spectral band selection. These

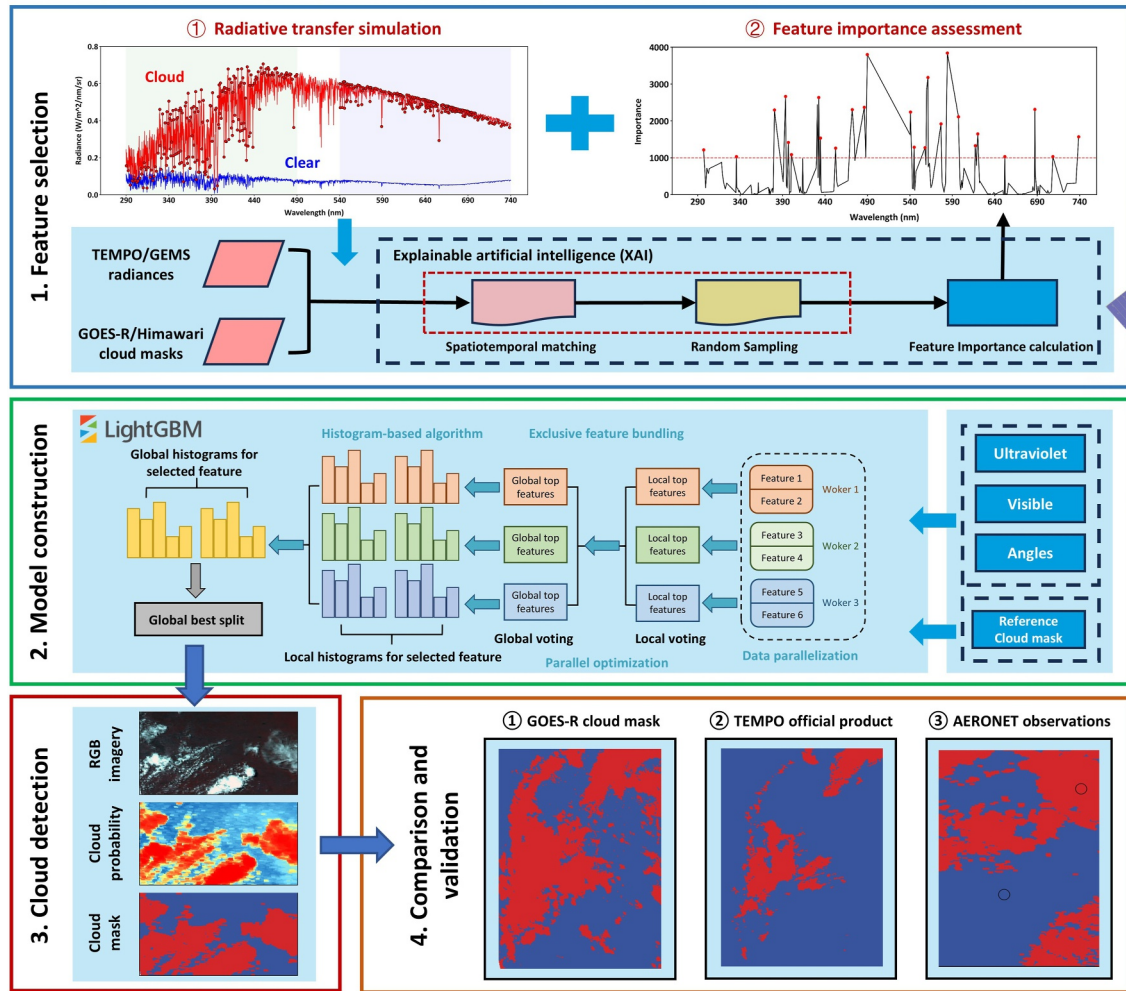


Figure 1. Flowchart of the Physics-Guided Feature Selection Machine Learning (PGML) cloud detection framework, illustrated using TEMPO observations.

analyses enabled a preliminary screening of cloud-sensitive spectral channels for dimensionality reduction, resulting in the selection of 400 bands.

To further refine the selection and determine the final set of input bands, an eXplainable artificial intelligence (XAI) approach (Arrieta et al., 2020; Gunning et al., 2019) was employed, using feature importance analysis to rank the relevance of individual spectral channels. By combining the flexibility of decision trees with the boosting framework, gradient boosting algorithms achieve high predictive accuracy and robustness, making them widely used for classification and regression tasks (Friedman, 2001). LightGBM (Ke et al., 2017) computes feature importance values during tree construction, enabling efficient identification of the most influential spectral channels. Formally, let the data set be $D = \{(x_i, y_i)\}_{i=1}^N$, where x_i represents the spectral feature vector and y_i indicates the cloud label. At each split node t , the algorithm evaluates candidate features f to maximize the Information Gain (IG) (Kent, 1983) defined as

$$IG(D, f) = H(D) - \sum_{v \in V(f)} \frac{|D_v|}{|D|} H(D_v) \quad (1)$$

where $H(D)$ denotes the impurity of data set D , and D_v represents the subset of D where feature f takes value v . The impurity measure H can be expressed in terms of entropy or Gini impurity, depending on the objective function. In LightGBM, impurity reduction is often computed using the Gini index (Gastwirth, 1972), which is defined as

$$\text{Gini}(D) = 1 - \sum_{c=1}^C p(c|D)^2 \quad (2)$$

where $p(c|D)$ is the proportion of samples belonging to class c within data set D , and C denotes the number of classes. The overall feature importance score $I(f)$ for feature f is then calculated by accumulating the impurity reduction across all trees and all nodes where feature f is used for splitting:

$$I(f) = \sum_{t \in T} \Delta G(t, f) \quad (3)$$

where T denotes the set of all trees in the ensemble, and $\Delta G(t, f)$ represents the gain in purity at node t when splitting on feature f . For regression tasks, the gain is computed based on the reduction in mean squared error (MSE) (Chicco et al., 2021).

We did not apply principal component analysis (PCA), independent component analysis (ICA), or autoencoder-based dimensionality reduction because these methods require processing the full hyperspectral data set simultaneously and produce latent features that are difficult to interpret physically. Instead, we used an XAI-based feature importance approach to directly identify the most informative spectral bands, providing an efficient and physically interpretable approach for dimensionality reduction.

After model training, the built-in feature importance values were extracted from LightGBM using the attribute model, representing the cumulative gain for each spectral band or derived feature. These values were exported for further analysis to evaluate the relative contribution of each feature in distinguishing cloudy and cloud-free samples. Based on the resulting importance rankings, bands with importance scores above 1000 were retained as the final inputs, yielding 6 UV bands and 19 VIS bands. While we acknowledge that extracting built-in Gini/MSE values alone is a relatively superficial application of XAI, the 25 selected wavelengths are strategically distributed across key spectral regimes that align with established radiative transfer principles for cloud detection: the ultraviolet region (<400 nm) leverages Rayleigh scattering sensitivity to detect thin clouds; the blue to green region (400–550 nm) exploits multiple scattering to differentiate clouds from low-reflectance land surfaces; and the red to near-visible region (>600 nm) provides strong contrast against vegetated surfaces due to the high, uniform reflectance of clouds across the visible spectrum versus wavelength-dependent surface absorption (He et al., 2021; Platnick et al., 2003). This physically grounded spectral mapping demonstrates that the model-selected features are informed by atmospheric scattering properties and absorption depths rather than purely statistical correlations.

Our motivation for pre-training dimensionality reduction is that TEMPO contains thousands of highly spectrally resolved bands, many of which are redundant or noisy for the specific task of cloud detection. Our feature selection approach identifies the most informative bands that are sensitive to cloud detection, thereby reducing computational overhead while maintaining or even improving accuracy, as demonstrated in previous studies. Thus, dimensionality reduction serves not only as a data compression strategy to improve computational efficiency, but also as a means to enhance model generalization by removing redundant or irrelevant spectral information.

3.1.2. Model Construction

In this study, LightGBM was selected as a lightweight model well-suited for large-scale, high-dimensional data sets. LightGBM is a highly efficient Gradient Boosting Decision Tree (GBDT) framework that incorporates Gradient-based One-Side Sampling and Exclusive Feature Bundling, introduced in 2017, enabling up to 20-fold acceleration over conventional GBDT with comparable accuracy to other models such as XGBoost (Ke et al., 2017). The model employs a histogram-based algorithm with exclusive feature bundling and parallel optimization, which enables the construction of both global and local histograms for the selected features, with the most relevant subsets iteratively identified through global voting, and multiple workers are assigned to process different feature bundles in parallel.

After spatiotemporal matching, TEMPO radiances were integrated with GOES-R cloud masks to construct the sample data sets. From these collocated pairs, approximately 4 million samples were randomly selected. The final

set of 25 TEMPO spectral bands (Table S1 in Supporting Information S1), together with the four geometric parameters (i.e., solar zenith angle, satellite zenith angle, relative azimuth angle, and scattering angle), was then used as input features to train the LightGBM model, with GOES-R cloud masks serving as supervisory labels. To ensure optimal performance, the model was trained using a cross-entropy loss function, which enhances its ability to learn fine-grained spectral distinctions between cloudy and clear-sky pixels, thereby improving discrimination capability and convergence stability. The final model parameters were determined by minimizing this loss across all training iterations.

During inference, the trained LightGBM model was applied to TEMPO L1B data, producing for each pixel a probabilistic estimate of cloud occurrence using the *predict_proba* function, which is a method of the LightGBM model (and other scikit-learn-style classifiers) that outputs the predicted probability of each class (ranging from 0 to 1). This output is interpreted as the cloud probability, representing the likelihood of cloud presence within a pixel and enabling more accurate characterization of sub-pixel cloud variability and transitional cloud conditions. A morphological operation with a 3×3 -pixel kernel was applied to refine the initial cloud probability retrievals, smoothing cloud boundaries and enhancing spatial coherence. Following the TROPOMI cloud-mask generation approach (Diego et al., 2021), a threshold of 0.5 was applied to the model-derived cloud probability to generate the final cloud mask. However, applying a fixed 0.5 threshold may be suboptimal across heterogeneous landscapes (e.g., urban, barren). A dynamic, surface-dependent thresholding strategy could potentially improve boundary detection, which will be investigated in future work. The retrieval results were comprehensively evaluated through intercomparison with GOES-R and TEMPO cloud mask products, as well as independent validation with AERONET observations.

3.2. Validation Approach

To comprehensively evaluate the performance of the proposed PGML method for cloud detection, this study employed seven evaluation indices, which could be calculated by following Equations 4–10. Specifically, we report Overall Accuracy (OA) and Balanced Overall Accuracy (BOA) (Skakun et al., 2022); both measure global classification correctness, but BOA additionally accounts for class imbalance.

$$OA = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

$$BOA = 0.5 * \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right) \quad (5)$$

where TP, TN, FP, and FN represent the true positive, true negative, false positive, and false negative, respectively (Wei et al., 2020). Furthermore, Recall quantifies detection sensitivity (how many true clouds are captured), while Precision measures prediction reliability (how many detected clouds are correct), and F1-score (F1) balances both:

$$Recall = \left(\frac{TP}{TP + FN} \right) \quad (6)$$

$$Precision = \left(\frac{TP}{TP + FP} \right) \quad (7)$$

$$F_1 = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (8)$$

Intersection over Union (IoU) offers a stricter, spatially sensitive evaluation than pixel-wise metrics like OA or F1, as it penalizes boundary mismatches and fragmented cloud patterns by measuring overlap relative to the union of detected and reference masks. The Cloud False Alarm Rate (FAR) was also introduced to quantitatively assess the associated false detection rate:

$$IoU = \frac{TP}{TP + FP + FN} \quad (9)$$

$$\text{FAR} = \left(\frac{\text{FP}}{\text{FP} + \text{TN}} \right) \quad (10)$$

In addition, we implemented a 10-fold cross-validation (CV) approach to rigorously evaluate model generalizability. Specifically, two strategies were adopted: temporal-based CV and spatial-based CV. Samples were randomly partitioned into 10 subsets according to time (hour) or space ($1^\circ \times 1^\circ$ grid). In each iteration, nine subsets were used for training and the remaining one for validation, with the process repeated 10 times so that each subset served exactly once as the validation set.

4. Results and Discussion

4.1. Cloud Probability Retrieval From TEMPO Imagery

Variations in illumination intensity influence the radiance received by satellite sensors; therefore, it is crucial to assess the performance of the PGML method under different illumination conditions. To this end, we present the spatial distribution of the TEMPO-retrieved results over North America during summer, covering hourly observations from 07:33 to 16:13 local time (Figure 2). The three primary columns present the full-scene (central) color-composite image (Channels: 2041-1039-1143), our cloud probability retrievals, and the official CLDO4 product. In the comparison maps, warmer colors (red to yellow) indicate higher cloud probabilities or fractions, while cooler colors (blue) indicate clearer skies. For thick clouds, our method yields cloud probabilities of ~ 0.9 , while the CLDO4 product estimates cloud fractions around the same level. Along cloud edges where clouds become optically thinner, CLDO4 typically reports cloud fractions of ~ 0.7 , whereas our retrieval remains higher. For thin and cirrus clouds, our method generally yields cloud probabilities around ~ 0.7 , while CLDO4 reports considerably lower fraction values (~ 0.5). Additionally, over the water body located southeast of the study domain (red rectangle in Figure 2), CLDO4 reports effective cloud fractions of approximately 0.5 for broken clouds, whereas our retrieval yields cloud probabilities exceeding 0.7. This is likely because, under these conditions, the actual cloud albedo is expected to be < 0.8 , which is lower than the CLDO4-assumed albedo of 0.8, leading to lower cloud fraction estimates.

To further examine temporal variability and spatial detail, zoom-in images for each time step are shown alongside the full-scene composites. The corresponding color composites illustrate how cloud appearance varies with varying solar illumination. During the early morning (07:33 local time) and late afternoon (16:13 local time), when the solar zenith angle is large and illumination is weak, shadows and contrast variations become more pronounced in the imagery. Temporal subsets highlight finer-scale cloud structures throughout the day and provide a more precise assessment of the cloud-detection performance. As fragmented clouds gradually dissipate, CLDO4 typically reports values of ~ 0.6 , whereas our retrieval yields slightly higher values, up to ~ 0.8 . Despite these differences in magnitude, both products capture highly consistent spatial cloud structures, particularly in the distribution and evolution of broken cloud features.

In the surrounding zoom-in panels, the diurnal evolution of cloud cover is more clearly illustrated. From 07:33 to 10:13 local time, clouds progressively increased in coverage and aggregation. During this period, CLDO4 generally reports cloud fraction values of ~ 0.7 near cloud edges, whereas our retrieval yields somewhat higher values, closer to 0.9. While a 3×3 spatial kernel is employed to enhance spatial consistency, the observed cloud-edge continuity stems predominantly from the model's spectral sensitivity to gradual cloud transitions rather than from the smoothing filter itself. However, we acknowledge that the kernel may introduce some degree of spatial smoothing, which could slightly expand cloud boundaries in certain scenes. From 11:13 to 16:13 local time, clouds thin and transition toward cirrus-like structures as they dissipate. Under these conditions, CLDO4 effective cloud fractions decrease to ~ 0.6 overall, dropping to ~ 0.4 – 0.5 in regions containing optically thin clouds, reflecting increased radiation transmission through partially transparent clouds. Meanwhile, our method produces cloud probabilities above 0.8 for these thin and semi-transparent clouds, reflecting the different physical meanings of the two products.

Having detailed the summer retrieval results, we extend the analysis to a winter case to evaluate the algorithm's performance under contrasting seasonal conditions (Figure 3). Despite lower solar elevation angles and reduced insolation in winter, apparent diurnal variation in the retrieved cloud probability remains evident. In optically thick cloud regions, our retrieval yields cloud probabilities above 0.8. The CLDO4 product reports effective cloud

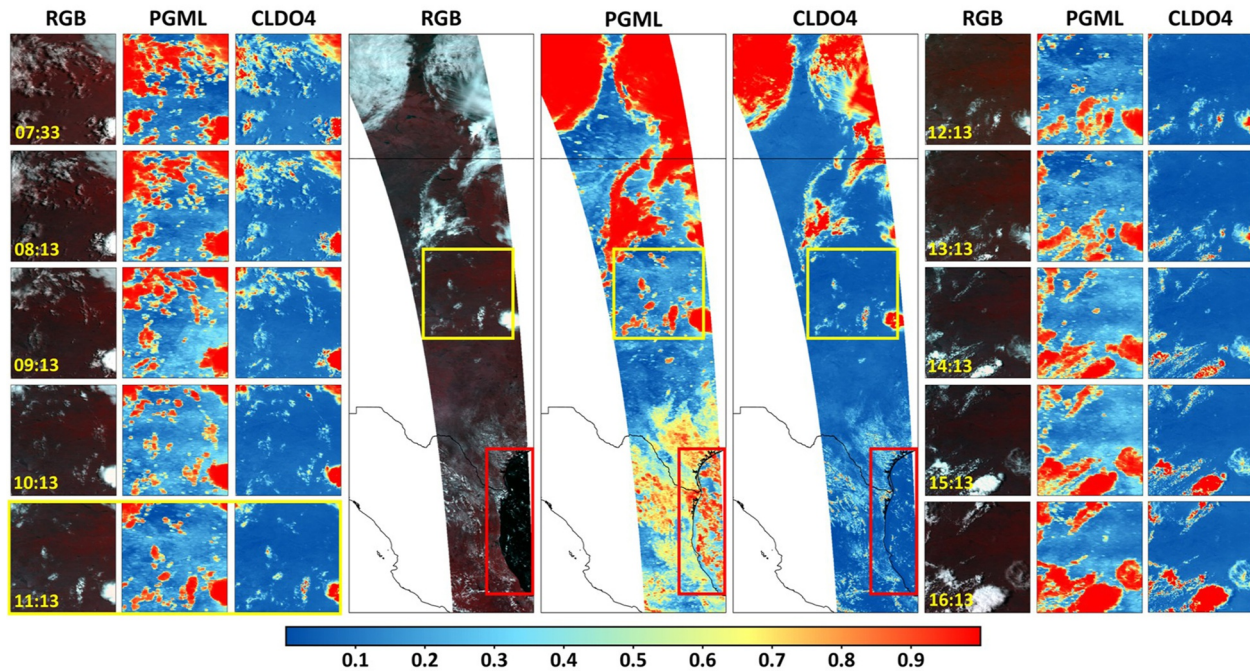


Figure 2. Diurnal evolution of TEMPO-retrieved results over North America on 25 August 2024 (Summer case). Shown are the full-scene (central) and zoom-in (surrounding) color-composite images (Channels: 2041-1039-1143), PGML retrievals, and the official CLDO4 product.

fractions of approximately 0.4 near cloud edges and around 0.5 for optically thin clouds, whereas our method produces cloud probabilities around 0.7. These differences are expected because optically thin or partially transmitting clouds allow greater radiation transmission, resulting in lower CLDO4 effective cloud fractions. A similar pattern is evident in the southern part of the scene, particularly over the ocean and coastal areas, where the official product reports cloud fractions below 0.3, while our method consistently reports probabilities above 0.7. Our PGML model performs generally well on saturated bright cloud scenes by leveraging multi-band spectral

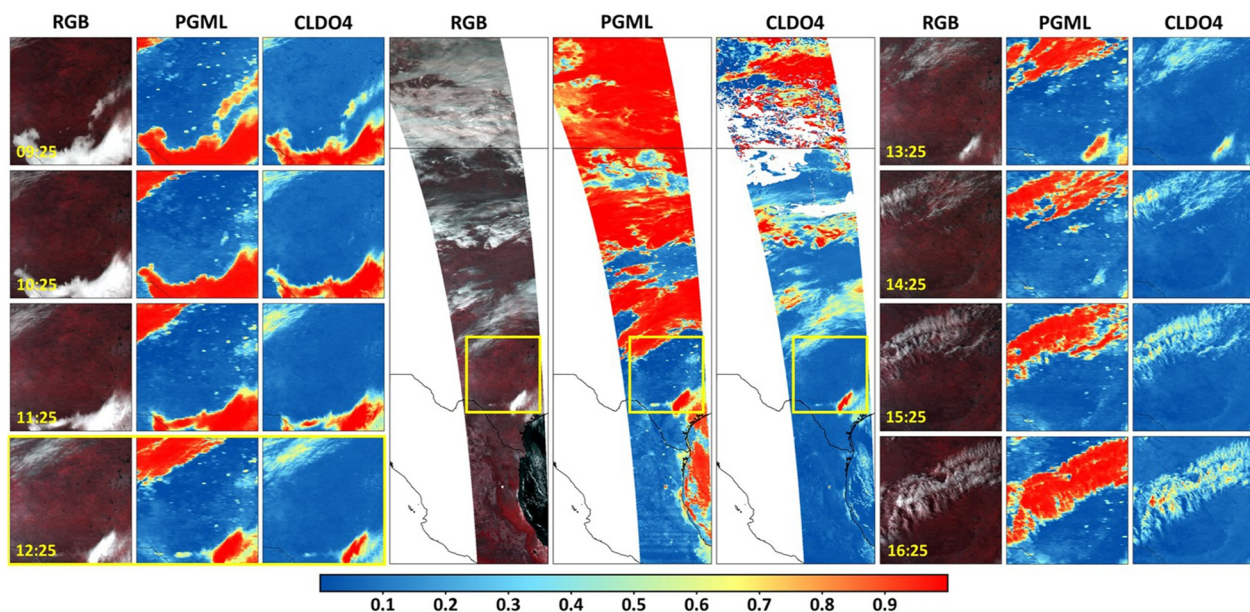


Figure 3. Diurnal evolution of TEMPO-retrieved cloud results over North America on 21 February 2024 (Winter case). Shown are the full-scene (central) and zoom-in (surrounding) color-composite images (Channels: 2041-1039-1143), PGML retrievals, and the official CLDO4 product.

feature learning, reliably classifying saturated, high-reflectance pixels as cloudy (probability ≈ 1.0) with few false negatives (e.g., Figure 3, top-left). Nevertheless, the inherent saturation issues of the TEMPO instrument represent a hardware limitation that may affect the confidence estimates of the model.

Zoom-in regions further highlight the temporal evolution of cloud properties throughout the day. At 09:25, PGML yields cloud probabilities above 0.9, while CLDO4 reports cloud fractions greater than 0.9. As the day progresses, the dense cloud deck gradually diminishes. Our retrieval consistently captures the reduction in high cloud-probability areas. By contrast, the official product exhibits a much sharper decrease in cloud fraction during this transition, particularly when the cloud field becomes more fragmented. Later in the afternoon, as the thick clouds disintegrate into broken cloud structures while still maintaining high cloudiness, our method remains sensitive to these morphological changes, producing stable estimates (>0.8). In contrast, the official product shows a sharp decline in cloud fraction from ~ 0.9 to <0.5 within an hour. However, the stationary speckles in our results likely occur over bright surfaces (e.g., urban or barren areas), where spectral signatures can partially overlap with those of thin clouds, thereby leading to mild false positives. Moreover, the use of a 3×3 spatial kernel may further enhance these speckle-like artifacts.

Seasonal cases for spring and autumn are provided in (Figures S1 and S2 of Supporting Information S1). Across seasons, our method consistently maintains stable performance in detecting thin, broken, and transitional clouds, as well as tracking diurnal changes. We further compare the cloud retrievals from the TEMPO spectral measurements using our method with those from CLDO4 over different surface types (Figure S3 in Supporting Information S1). Over water, both methods estimate thick clouds (~ 0.9) similarly, but CLDO4 generally reports lower effective cloud fractions (~ 0.6 – 0.7) for thin and edge clouds, reflecting increased radiation transmission through optically thin or partially cloudy regions, whereas our method provides higher and more spatially continuous cloud probabilities. Over vegetation, this difference becomes more pronounced: thick-cloud fractions in CLDO4 remain lower (~ 0.7) due to its radiative retrieval approach, and fragmented clouds appear scattered, while our method maintains more coherent cloud probability structures. Over barren surfaces, both methods perform comparably for thick clouds (~ 0.9). In urban regions, neither method shows obvious false detections; however, CLDO4 reports lower effective cloud fractions for thin clouds (~ 0.4) compared with our retrievals (~ 0.7), reflecting the different physical meanings of effective cloud fraction and cloud probability.

Also, we included a quantitative evaluation of our model over snow and ice using the MCD12C1 land-use-type data set. Our model maintains good performance in detecting clouds over snow and ice, with higher accuracy compared to the TEMPO L1B cloud mask (OA = 98.00% vs. 97.60%, BOA = 62.50% vs. 50.00%; Table S2 in Supporting Information S1). Although PGML achieves reasonable performance over these surfaces, bright snow- and ice-covered land can still be misclassified as cloud in some cases due to the lack of infrared (IR) information.

Compared with CLDO4, our method generally reports higher cloud retrieval values, whereas CLDO4 often yields lower values, particularly along cloud boundaries and within thin-cloud fields. These differences arise from the distinct retrieval strategies and design objectives of the two products: CLDO4 represents an effective cloud fraction optimized for TEMPO trace-gas retrievals in the UV band, emphasizing radiative consistency, and is therefore expected to be lower than a geometric cloud probability in many situations. In contrast, our product delivers a cloud probability derived from a machine-learning framework trained on GOES-R cloud masks, prioritizing cloud detection over radiative assumptions. This approach is highly useful for applications such as aerosol retrievals, particularly under broken or thin-cloud conditions. However, this GOES-R-like product may not be suitable for TEMPO UV trace-gas retrievals, because the light path and air-mass factor (AMF) calculations for UV trace gases require a UV-based cloud product. Visible or IR-based cloud products can provide complementary information but are not the primary inputs for operational UV trace-gas retrievals.

4.2. Evaluation of Generated Cloud Masks From TEMPO Imagery

4.2.1. Comparison With GOES-R Cloud Masks

Figures 4 and 5 compare cloud-mask results corresponding to Figures 2 and 3, derived from the GOES-R reference cloud mask, PGML retrievals, and the TEMPO L1B cloud mask. Overall, our method exhibits high consistency with GOES-R across thick, thin, and broken-cloud regions. By contrast, the TEMPO L1B product shows substantial omissions, especially along cloud edges, for thin or broken clouds, and for clouds over darker surfaces. In the southern domain, the TEMPO L1B cloud mask frequently identifies only the cores of thicker

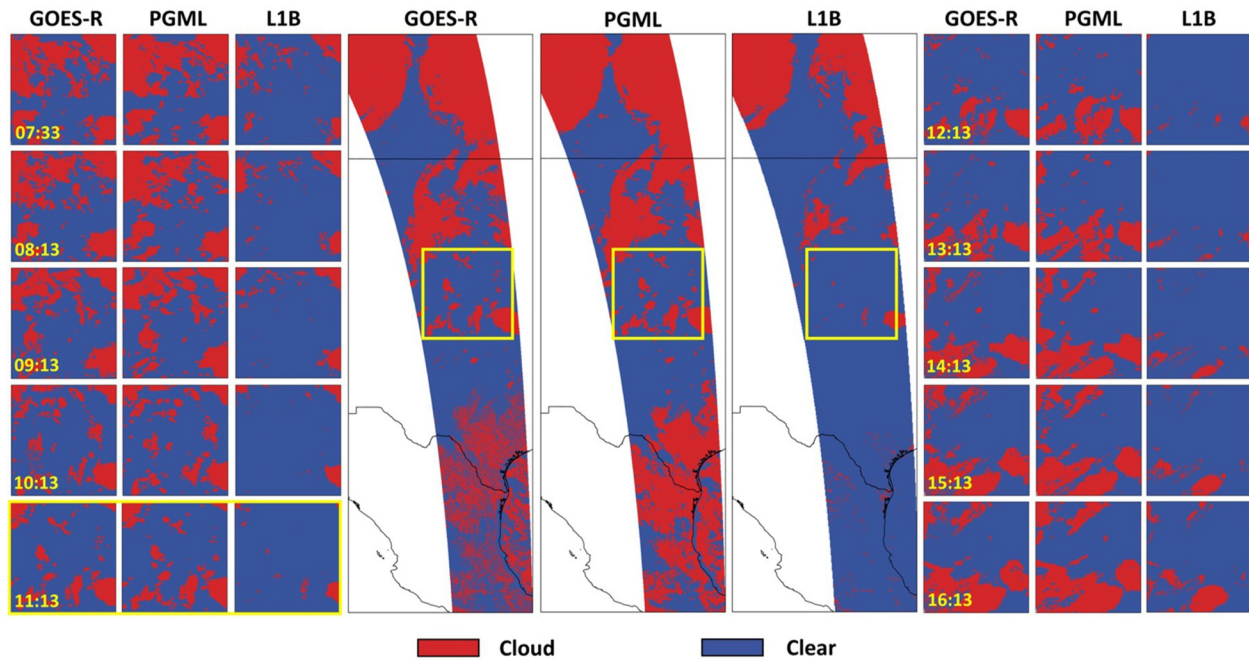


Figure 4. Comparison of the diurnal variations in cloud-mask results corresponding to Figure 2, derived from the GOES-R reference cloud mask, PGML retrievals, and the TEMPO L1B cloud mask over North America on 25 August 2024 (Summer case).

cloud patches, missing most broken clouds. From 07:33 to 09:13 local time, GOES-R and our algorithm detect extensive thin cloud cover in the upper-left region, whereas the TEMPO L1B cloud mask identifies only a small fraction of this cloud field. By 13:13 local time, GOES-R and our algorithm still detect remaining cloud patches, while the TEMPO L1B cloud mask classifies nearly all such areas as clear sky.

In Figure 5, differences become more pronounced: the L1B cloud mask performs comparably to GOES-R only in the thick-cloud region but significantly underestimates cloudiness elsewhere, including broken clouds over the ocean. From 09:25 to 12:25 local time, all three results capture the gradual dissipation of the thick cloud in the

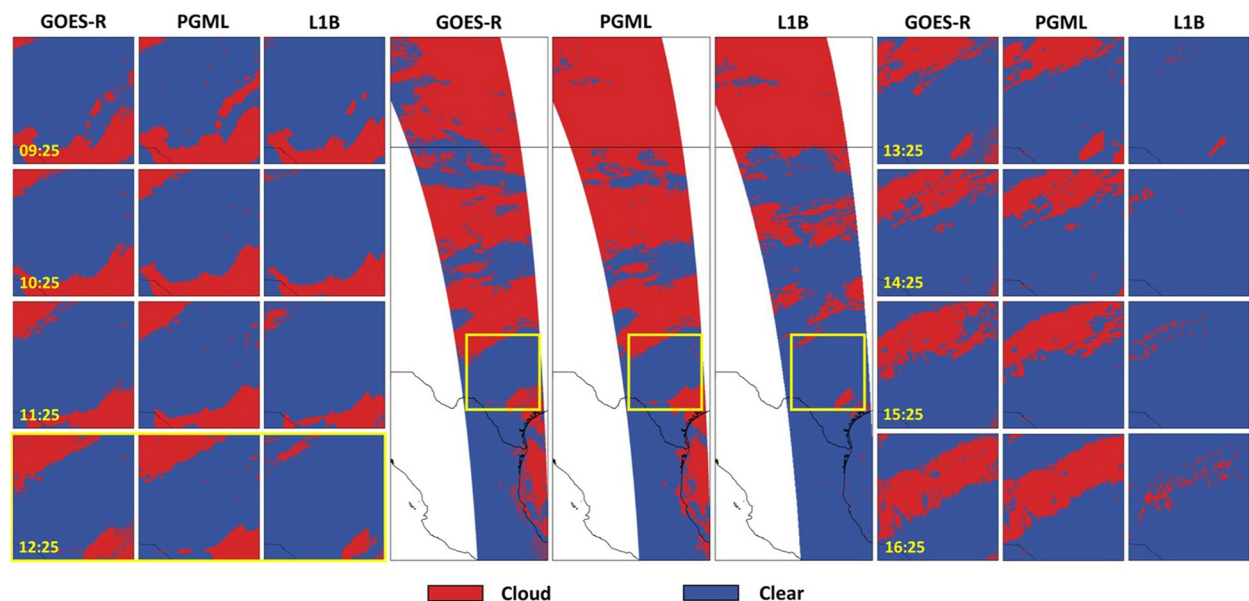


Figure 5. Comparison of the diurnal variations in cloud-mask results corresponding to Figure 3, derived from the GOES-R reference cloud mask, PGML retrievals, and the TEMPO L1B cloud mask over North America on 21 February 2024 (Winter case).

lower part of the scene, indicating generally consistent performance during this period. However, from 13:25 to 16:25 local time, GOES-R and our method continuously track a cloud patch migrating from the upper-left region toward the scene center. In contrast, the L1B product fails to detect this cloud for most hours and only partially identifies it at 16:25, resulting in a substantial loss of cloud information.

Additional examples in Spring and Autumn cases (Figures S4 and S5 in Supporting Information S1) further support our findings. Both GOES-R and our method successfully detect cirrus clouds, while the L1B cloud mask product misclassifies them as clear sky; our method also preserves spatial continuity along thick-cloud boundaries and aligns closely with GOES-R, while the L1B product often omits parts of cloudy edge regions. This systematic underestimation is likely due to the limited spectral information used in the operational approach, which reduces sensitivity to optically thin clouds.

We further compare cloud masks across diverse surface types using the GOES-R cloud mask as the reference (Figure S6 in Supporting Information S1). Over water, the TEMPO L1B mask shows low detectability for thin clouds, often confusing them with the underlying surface. In contrast, our method aligns more closely with GOES-R, though slight differences occur at cloud boundaries. Over vegetation, performance is broadly similar to GOES-R, but our method provides more spatially continuous detection for broken-cloud fields, likely due to the dilation processing applied in our model, which smooths cloud boundaries and provides a more cohesive representation. The L1B algorithm performs poorly and fails to detect significant clouds over bright surfaces (see Figure S3 in Supporting Information S1), such as barren land and urban areas. In particular, for thin clouds above urban areas, the L1B product fails to detect any cloud signal in the first case. This under-detection is likely caused by the low reflectance contrast between thin clouds and high-albedo surfaces, which complicates cloud-surface discrimination. Overall, comparisons against GOES-R demonstrate that our method successfully captures thin clouds over bright urban surfaces and shows strong robustness across diverse surface conditions.

Figure 6 shows a radar chart illustrating the hourly performance of our cloud mask relative to the GOES-R reference mask, using seven evaluation indices (OA, BOA, F_1 , Recall, Precision, IoU, and 1-FAR). For (1-FAR), we use it solely to maintain a consistent visualization convention in which higher values indicate better cloud detection performance. Our method demonstrates consistently strong performance throughout the day, with BOA values remaining above 82% for most hours, substantially higher than the ~73%–75% achieved by the L1B product. During the morning, when solar illumination is weak and cloud-surface radiance contrast is low, our model maintains robust performance, with BOA values of ~87% at 07:00 and 08:00, compared with ~75% from the L1B algorithm. From 09:00 to 13:00, despite rapid changes in solar illumination, our method maintains high accuracy (BOA = 82%–84%), consistently outperforming the L1B product by 8%–10%. In the afternoon (14:00–16:00), when shadows and cloud anisotropy become more pronounced, our method continues to perform reliably (BOA = ~84%), while the L1B product declines to 71%–74%. Overall, our model achieves OA = 86.44% and BOA = 84.43%, compared with OA = 71.67% and BOA = 73.86% for the L1B product. Although the L1B product attains slightly higher precision (94.66% vs. 84.95%), our model achieves a dramatically higher recall (93.85% vs. 51.28%), highlighting the L1B product's tendency to misclassify cloud pixels as clear. In contrast, our approach provides more accurate and robust cloud detection under varying solar illumination and cloud conditions throughout the day. In addition, the spatial-based cross-validation achieved consistently high accuracy (average OA = 84.41%, BOA = 83.32%, Table S3 in Supporting Information S1), while the temporal-based cross-validation showed similar performance (OA = 82.96%, BOA = 82.36%, Table S4 in Supporting Information S1), further supporting the model's generalizability.

Additionally, we also compared our cloud masks with a binary cloud mask derived from the TEMPO CLDO4 product using a cloud fraction threshold of 0.1 (Figure S7 in Supporting Information S1). The results show that, when evaluated against the GOES-R reference masks, this threshold-based CLDO4 mask achieves a BOA of 75.79% and an OA of 78.32%. These values are highly comparable to those of the TEMPO L1b cloud mask (BOA = 73.86%, OA = 71.67%). However, both masks perform substantially worse than our proposed PGML model (BOA = 84.43%, OA = 86.44%), illustrating the robustness of our approach.

4.2.2. Validation With AERONET Observations

We further evaluate our cloud detection results through independent comparisons with AERONET observations spanning various surface types and cloud conditions. The three cases (Figure 7) demonstrate that both the centers and boundaries of thick clouds identified by AERONET are accurately captured by our algorithm. In regions

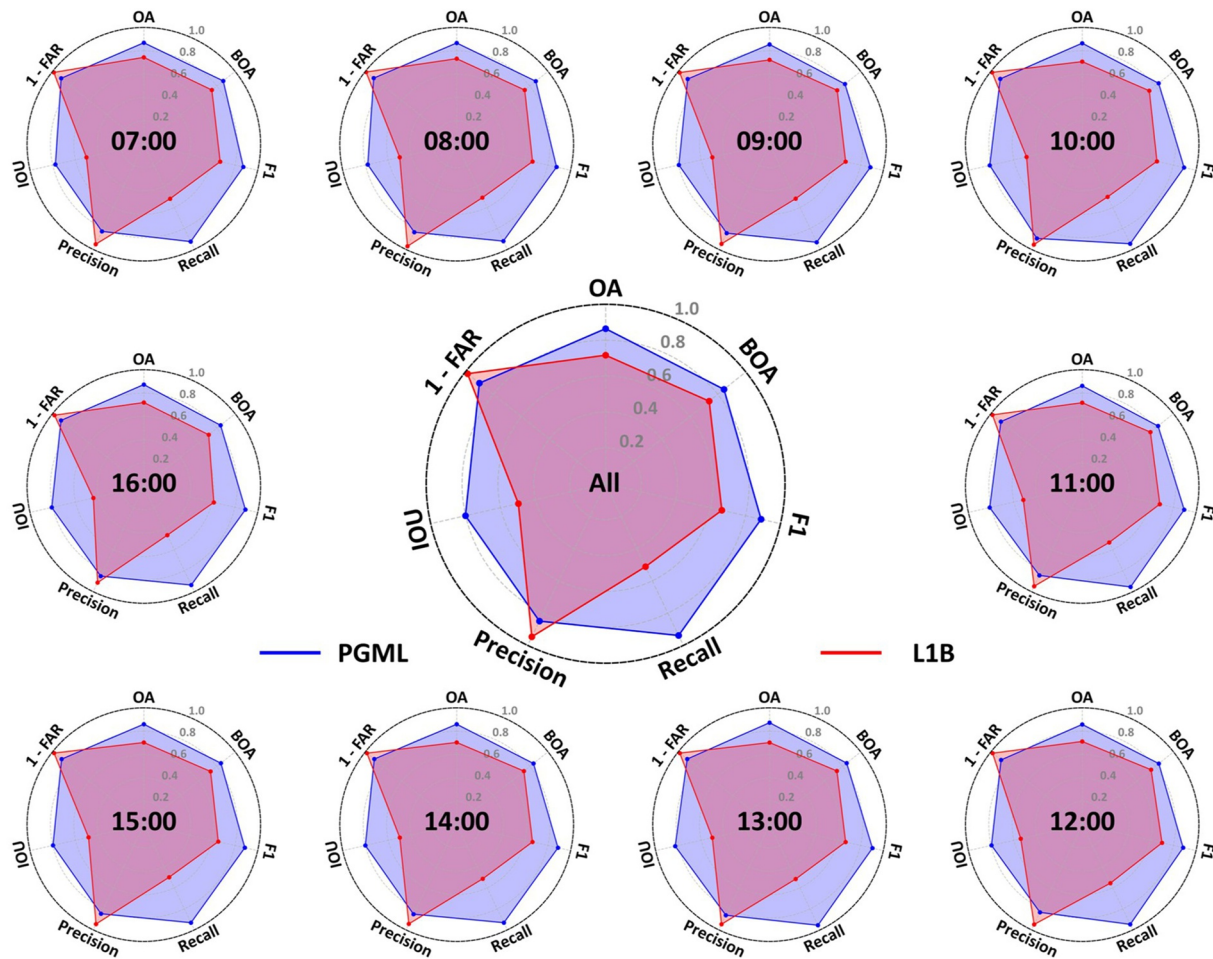


Figure 6. Hourly performance comparison between PGML cloud mask (blue lines) and the TEMPO L1B cloud mask (red lines), using the GOES-R cloud mask as the reference.

classified as clear in the color-composite imagery, AERONET observations confirm that our method does not introduce false cloud detections. Another three cases (Figure S8 in Supporting Information S1) further highlight the algorithm's ability to detect thin clouds. For clouds with TEMPO-retrieved probabilities of approximately 0.6–0.7, AERONET classifies these regions as cloudy, indicating that our method does not underestimate cloudiness. Moreover, in the third row, narrow clear-sky gaps between adjacent thin-cloud patches are well preserved in our retrievals, consistent with the AERONET records. This highlights the model's ability to maintain fine spatial detail without over-smoothing cloud boundaries. Compared with the L1B cloud mask, our algorithm captures a substantially more complete cloud cover, whereas the L1B mask shows noticeable missed detections, particularly for thin and broken clouds. Furthermore, variations in the hourly Cloud Occurrence Frequency (defined as the ratio of cloudy samples to total samples within each hourly bin, Figure S9 in Supporting Information S1) demonstrate that our model more effectively captures diurnal cloud probability variability, achieving a higher correlation with AERONET observations ($R = 0.621$) than the TEMPO L1B cloud mask ($R = 0.479$).

Overall, our method exhibits greater stability and substantially higher accuracy than the L1B cloud mask product (OA: 77.53% vs. 68.00%; BOA: 78.54% vs. 54.88%) (Figure 8a). The F_1 score shows an even more pronounced, nearly threefold improvement (68.09% vs. 23.40%), indicating that the L1B product struggles to correctly identify clouds, especially thin or semi-transparent cloud layers. A similar pattern is observed for the IoU, with our method attaining 51.61% compared to only 13.25% for the L1B product, highlighting the latter's difficulty in producing spatially consistent cloud detection. While precision is comparable between the two methods (58.72% vs. 61.11%), our algorithm exhibits a markedly higher cloud hit rate (81.01% vs. 14.47%), consistent with the benchmarks against GOES-R. These findings are further supported by the high sensitivity of AERONET

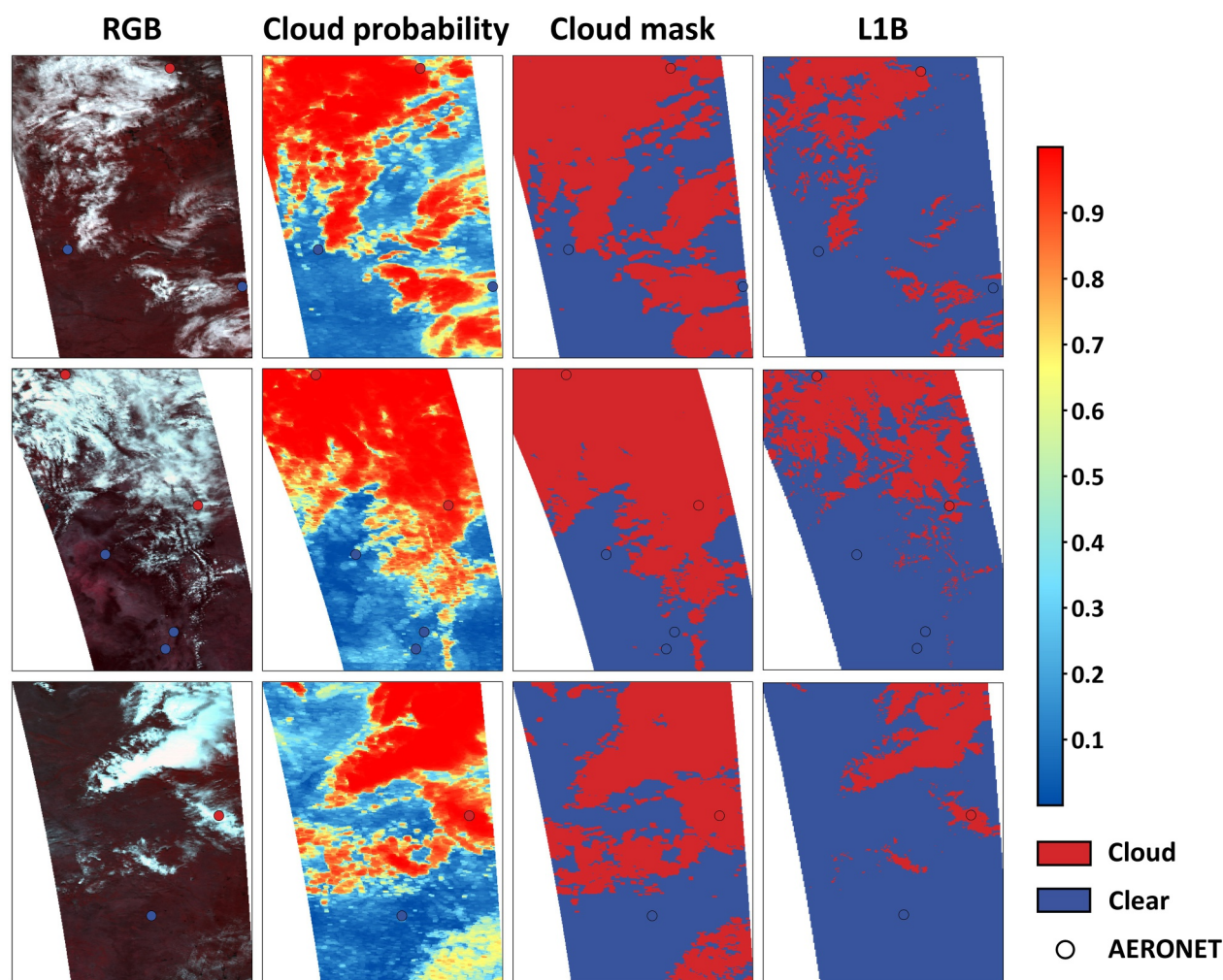


Figure 7. Spatial distribution of our TEMPO-retrieved cloud probability and cloud mask, compared with the TEMPO L1B cloud mask, using AERONET observations as reference.

observations to optically thin clouds. The L1B product frequently misses such clouds, whereas ours successfully captures them, contributing to its superior performance.

4.3. Evaluation of Model Generalizability on GEMS Imagery

To evaluate the adaptability and transferability of our model, we extend it to GEMS observations. Like TEMPO, GEMS is a hyperspectral UV-visible imager in geostationary orbit onboard the Geostationary Korea Multi-Purpose Satellite 2B (GEO-KOMPSAT-2B) (Kim et al., 2020), observing atmospheric composition over Asia with a spectral range of 300–500 nm and a spectral resolution of 0.6 nm (Shu et al., 2023). The Himawari-8 official cloud mask product was used as the reference label for constructing the cloud detection model, with 1 million samples randomly selected. Using the same framework as for TEMPO, we finally selected 22 spectral bands from UV to visible wavelengths for inputs (Table S5 in Supporting Information S1). We did not attempt to directly apply the same 25 wavelengths selected from TEMPO to GEMS for two main reasons. First, GEMS has a narrower spectral range (300–500 nm) compared to TEMPO (290–740 nm). Consequently, many of the wavelengths selected for TEMPO—particularly those in the red and near-visible region (>600 nm)—fall outside GEMS's spectral coverage and are therefore unavailable. Second, while a fully transferable wavelength set across instruments would be ideal, we suspect that the optimal wavelength subset may vary depending on each instrument's specific spectral response functions, signal-to-noise ratios, and calibration characteristics. Nevertheless, we agree that investigating cross-instrument generalizability is an important direction for future work. A

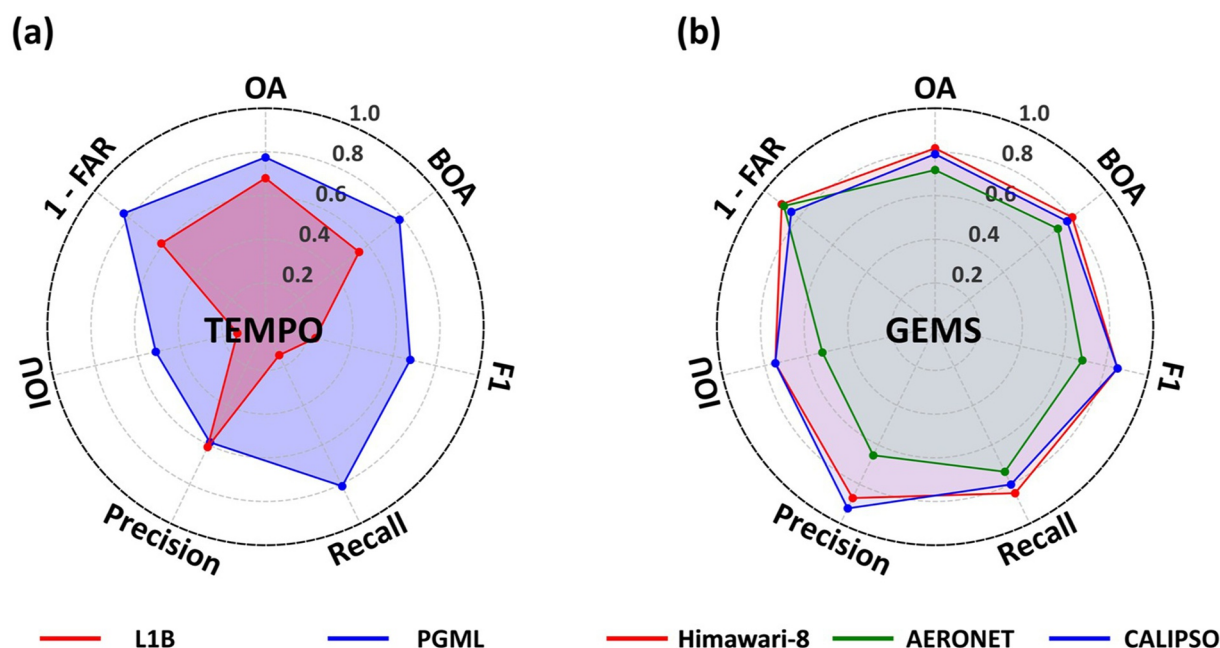


Figure 8. Validation and comparison of cloud detection performance: (a) TEMPO cloud detection evaluated against AERONET observations, comparing the PGML algorithm (blue lines) with the L1B cloud mask (red lines). (b) GEMS cloud detection evaluated against Himawari-8 cloud mask (red lines), AERONET (green lines), and CALIPSO (blue lines) observations.

systematic comparison using overlapping spectral windows (e.g., UV and blue bands common to both instruments) would be a logical next step to assess how well a model trained on TEMPO can be transferred to GEMS or other geostationary UV-VIS instruments.

Figure S10 in Supporting Information S1 illustrates the diurnal evolution of GEMS-retrieved cloud fraction and mask, compared with the Himawari-8 cloud mask over Asia. Our algorithm performs well across different cloud regimes: thick, continuous clouds reach fractions of ~ 0.9 , while thin and broken clouds remain around 0.6 – 0.7 , demonstrating good sensitivity to subtle radiative features. At 02:45, 05:45, and 06:45 UTC, thick cloud decks and coastal areas are well captured and distinguished, with cloud fractions above 0.9 , and masks closely matching the Himawari-8 reference. At 03:45 UTC, small broken cloud patches are detected that Himawari-8 misses, likely due to the model's enhanced sampling and strong feature-learning capacity. Even in fragmented clouds (e.g., 07:45 UTC), cloud fractions remain stable (~ 0.8), and cloud masks remain consistent with Himawari-8. Overall, cloud detection over ocean surfaces is robust, effectively capturing both thick and thin clouds with high spatial and temporal consistency throughout the diurnal cycle, and further demonstrating the model's adaptability to varying lighting and viewing angles.

For extensive thick clouds, our cloud fraction estimates (>0.8) and cloud masks agree well with both AERONET and CALIPSO observations (Figures S11 and S12 in Supporting Information S1). For thinner clouds with estimated cloud fraction ~ 0.6 , which are not easily distinguishable in the color-composite imagery, both our method and AERONET consistently detect their presence. Over coastal regions, transitions between cloudy and clear-sky conditions are clearly captured, with cloud fractions above 0.9 in cloud cores and around 0.8 along boundaries, consistent with CALIPSO observations. In ocean regions with scattered thin clouds, our cloud fractions range from 0.6 to 0.7 , and the resulting masks align well with CALIPSO. Among coastal areas, our method effectively distinguishes thin clouds from adjacent clear sky, demonstrating robust performance across diverse cloud types and surface conditions.

Overall, our cloud retrievals show high consistency with the Himawari-8 reference cloud mask, ground-based AERONET measurements, and CALIPSO overpass observations (Figure 8b). The OA and BOA values reach 81.67% and 80.38% for Himawari-8, 71.75% and 71.98% for AERONET, and 79.01% and 77.42% for CALIPSO, demonstrating strong generalization capability across different validation sources and viewing geometries. By using radiance inputs and incorporating physical constraints during feature selection, the model achieves broad

adaptability across hyperspectral geostationary sensors while retaining interpretability and reduced sensitivity to radiometric calibration uncertainties.

5. Conclusions

Accurate cloud prediction is crucial for improving the retrieval of both atmospheric and surface parameters from satellite observations. For newly launched satellites such as TEMPO, limitations in spectral coverage and calibration uncertainties complicate the application of conventional cloud-detection approaches. As a hyperspectral sensor, selecting appropriate channels and reducing spectral dimensionality remain long-standing challenges. In this study, we address these challenges by developing a Physics-Guided Feature Selection Machine Learning (PGML) framework that integrates radiative transfer simulations to guide spectral feature selection and eXplainable AI (XAI) to reduce dimensionality while preserving interpretability. A LightGBM model was trained using GOES-R cloud masks and then applied to TEMPO hyperspectral observations, and a similarly structured model was developed for GEMS using Himawari-8 as the reference.

Comparison results show that the PGML model estimates higher cloud probabilities and effectively captures diurnal variations, whereas the TEMPO CLDO4 product generally reports lower effective cloud fractions, as expected due to its radiative definition and the use of a cloud albedo of 0.8 for consistency with trace-gas retrievals. These differences reflect the distinct physical meanings of the two products rather than retrieval biases. The PGML cloud probability may provide additional information for broader applications, including aerosol retrievals. Compared with the TEMPO L1B cloud mask product, our method achieved substantial improvements in overall accuracy (86.44% vs. 71.67%) and especially in cloud hit rate (93.85% vs. 51.28%) against the GOES-R reference cloud mask. Independent validation with AERONET further confirmed these gains (OA: 77.53% vs. 68.00%). Our model also successfully adapts to GEMS observations, achieving OA of 81.67%, 71.75%, and 79.01% relative to the Himawari-8 reference mask, AERONET, and CALIPSO observations. These results demonstrate the robustness of the proposed model across geostationary hyperspectral sensors, surface types, and cloud conditions, highlighting its strong potential for operational implementation.

GOES-R provides a geometric cloud mask (binary cloud/no-cloud based on sub-pixel spatial uniformity), while trace-gas-based cloud fractions (e.g., from O_4 or O_2A band) represent an optically weighted effective cloud fraction sensitive to cloud optical depth and photon path length. Consequently, direct pixel-wise comparison between our model output (trained on GOES-R) and trace-gas cloud fractions is not intended to imply equivalence. Instead, we use trace-gas cloud fractions as an independent, physically distinct reference to assess whether our model captures optically thin clouds consistently. The systematic differences observed (e.g., our model showing higher cloud probability than O_4 -retrieved cloud fraction over broken cloud scenes) are expected and discussed as resulting from these fundamental physical discrepancies. The primary intended application of our PGML cloud mask is cloud screening for TEMPO aerosol retrievals.

However, several limitations should be acknowledged. Training on GOES-R masks inevitably caps the model's theoretical ceiling due to inherent label noise. Moreover, the lack of parallax correction for different viewing geometries may introduce minor geometric misalignments, particularly for high-altitude clouds. Future work could explore self-supervised or weakly supervised learning to reduce dependency on noisy labels, as well as incorporate parallax correction for applications that demand higher geometric fidelity.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The TEMPO Level 1B radiance and CLDO4 products are available from the NASA Atmospheric Science Data Center (2024a, 2024b). The CALIPSO Level 2 Vertical Feature Mask product is available from the NASA Atmospheric Science Data Center (2024c). The GOES-R CMa and Himawari-8 cloud products are available from the ICARE Data and Services Center (2018a, 2018b). The AERONET Level 1.0 and Level 1.5 data are available from the NASA Goddard Space Flight Center (2024). The GEMS Level-1 data are available upon request from the National Institute of Environmental Research Environmental Satellite Center (2024).

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